Components - specific troubleshooting practical

1) Open your desktop or laptop, locate the RAM slots, and remove and re-insert the RAM module to simulate reseating — then document each step you took and any issues you observed during the process.

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1. RAM Reseating – Documentation Steps

Performed:

- Shut down the PC completely.
- Switched off the power supply and unplugged the power cable.
- Pressed the power button for a few seconds to discharge remaining power.
- Opened the side panel of the computer case.
- Located the RAM slots on the motherboard.
- Released the retaining clips on both sides of the RAM slot.
- Carefully removed the RAM module.
- Checked the RAM contacts and slot for dust or damage.
- Reinserted the RAM module, ensuring the notch matched the slot.
- Pressed firmly until both retaining clips locked into place.
- Closed the case, connected the power cable, and started the PC.
- Verified in BIOS and Windows that the full RAM capacity was detected.

Issues Observed:

- No physical damage or dust was found.
- RAM was detected correctly after reseating.
- The PC booted normally without any errors.

2) Create a troubleshooting checklist for diagnosing a faulty GPU in a gaming PC, including at least 5 steps from initial symptom identification to confirming the fix.Hint: Think about display issues, driver checks, and physical connections.

Step	Action	Expected Result
1	Check for display issues (blank screen, artifacts, flickering).	Identify the symptom.
2	Verify monitor cable and GPU power connectors are securely connected.	Rule out loose connections.
3	Reseat the graphics card in the PCIe slot.	Ensure proper physical connection.
4	Update or reinstall the graphics driver.	Fix driver-related issues.
5	Check GPU temperature and cooling fans.	Detect overheating problems.
6	Test the GPU in another PC or install a known-good GPU.	Confirm whether the GPU is faulty.
7	Run a graphics stress test and monitor stability.	Verify the issue is resolved.

2) Simulate a storage issue by unplugging and replugging a SATA or NVMe drive (SSD/HDD) on a test system, then describe the error messages or BIOS indicators you see before and after reconnecting.

Before Reconnecting

- SATA SSD/HDD cable disconnected.
- BIOS message:

  No Boot Device Found

  Boot Device Not Detected
- Windows could not start because the storage device was not detected.

After Reconnecting

- Reconnected the SATA/NVMe drive securely.
- BIOS detected the SSD/HDD correctly.
- Windows booted normally.
- Storage device appeared in BIOS and File Explorer.

4) Given a scenario where a PC randomly restarts, list which component (RAM, GPU, storage) you would check first and explain your reasoning, then outline the specific diagnostic steps you would perform for that component.

Component to Check First: RAM Reason:

Faulty or improperly seated RAM is a common cause of random restarts, crashes, and blue screen errors. It can make the system unstable even when other hardware is functioning normally.

Diagnostic Steps

1. Power off the PC and disconnect the power cable.
2. Remove and reseat the RAM module.
3. Clean dust from the RAM slots if necessary.
4. Test one RAM stick at a time if multiple modules are installed.
5. Run **Windows Memory Diagnostic** or **MemTest86** to check for memory errors.
6. Verify BIOS detects the correct amount of RAM.
7. Use the PC for some time to confirm that random restarts no longer occur